

H E S T O N (1 9 9 3)

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1 Partial Differential Equation

Suppose the underlying asset value follows the diffusion:

$$dS_t = \alpha S_t dt + \sqrt{v_t} S_t dW_t^{S, \mathbb{P}}$$

where α is the mean growth rate, and $W_t^{\mathbb{P}}$ is the Brownian motion under the physical probability measure, and v_t is the variance following Cox-Ingersoll-Ross (1985) square-root process:

$$dv_t = \kappa(\bar{v} - v_t)dt + \xi\sqrt{v_t}dW_t^{v, \mathbb{P}}$$

with κ being the speed of mean reversion, $\bar{v}$ long-run average variance, and ξ is the volatility of variance parameter. $W_t^{S, \mathbb{P}}$ and $W_t^{v, \mathbb{P}}$ are two brownian motions driving innovations, such that $\mathbb{E}^{\mathbb{P}}[dW_t^{S, \mathbb{P}} dW_t^{v, \mathbb{P}}] = \rho dt$, with ρ , normally, being less than zero¹. Intuitively, κ controls variance persistence, while ρ and ξ skewness and kurtosis, respectively². Then suppose we have an option written on that asset, $V(S, v, t)$, and I construct a riskless portfolio of this option, Δ units of S_t , and ν units of another option $U(S, v, t)$ ³:

$$\Pi_t = V(S, v_t, t) + \Delta S_t + \nu U(S, v_t, t)$$

$$d\Pi_t = dV_t + \Delta dS_t + \Delta \delta S_t + \nu dU_t$$

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¹See, e.g. Black (1976) for a note on so-called “leverage effect.”

²Setting $\kappa = \xi = 0$ will bring us to the Black and Scholes (1973).

³Heston (1993) introduces market incompleteness – variance is not traded, we thus require another variance-exposed instrument to hedge it.

where δ is the return shortfall⁴. By Itô's Lemma

$$\begin{aligned} d\Pi_t = & \left(\frac{\partial V}{\partial t} + \frac{1}{2}v_t S_t^2 \frac{\partial^2 V}{\partial S^2} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 V}{\partial v^2} \xi \rho S_t v_t \frac{\partial^2 V}{\partial S \partial v} \right) dt \\ & + \nu \left(\frac{\partial U}{\partial t} + \frac{1}{2}v_t S_t^2 \frac{\partial^2 U}{\partial S^2} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 U}{\partial v^2} \xi \rho S_t v_t \frac{\partial^2 U}{\partial S \partial v} \right) dt \\ & + \left(\frac{\partial V}{\partial S} + \nu \frac{\partial U}{\partial S} + \Delta \right) dS_t + \left(\frac{\partial V}{\partial v} + \nu \frac{\partial U}{\partial v} \right) dv_t + \Delta \delta S_t dt \end{aligned} \quad (1)$$

which is riskless when the random terms cancel out, i.e. when portfolio dynamic is deterministic. With simple algebra I get:

$$\begin{aligned} \Delta &= -\frac{\partial V}{\partial S} - \nu \frac{\partial U}{\partial S} \\ \nu &= -\frac{\partial V}{\partial v} / \frac{\partial U}{\partial v} \end{aligned}$$

so that the last two terms in the Equation (1) disappear, and the portfolio should now earn the risk-free rate of return. Let me express equation (1) as

$$\begin{aligned} A + \nu B &= r(V_t + \Delta S_t + \nu U_t)dt \\ \frac{A - rV_t + rS_t \frac{\partial V}{\partial S}}{\frac{\partial V}{\partial v}} &= \frac{B - rU_t + rS_t \frac{\partial U}{\partial S}}{\frac{\partial U}{\partial v}} \end{aligned}$$

with r being a continuously compounded risk-free rate. LHS (RHS) side is simply function of V (U) only. Heston (1993) suggests writing them as the function of S , v , and t :

$$g(S, v, t) = -\kappa(\bar{v} - v_t) + \lambda(S, v, t) \quad (2)$$

where $\lambda(S, v, t)$ is the variance risk premium. Although the step may seem somewhat not intuitive, I attempt to justify it separately in the Section 5 below. Substitute this as the RHS to get

$$rV = \frac{\partial V}{\partial t} + (r - \delta)S_t \frac{\partial V}{\partial S} + \frac{1}{2}v_t S_t^2 \frac{\partial^2 V}{\partial S^2} + (\kappa(\bar{v} - v_t) - \lambda v_t) \frac{\partial V}{\partial v} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 V}{\partial v^2} + \xi \rho v_t S_t \frac{\partial^2 V}{\partial S \partial v} \quad (3)$$

where I followed Heston (1993) in assuming $\lambda(S, v, t) = \lambda v_t$, i.e. that the variance risk premium is linear in variance. If the variance becomes constant, the PDE will be reduced to the Black and Scholes (1973) PDE –

⁴So that the total expected return on the underlying is $\mu = \alpha + \delta$ (e.g. capital gain rate plus dividend yield for stocks). By holding the underlying, we receive the “convenience” yield, while by shorting we are required to pay it (otherwise, no rational investor would enter the corresponding long position).

three first terms.

Equation (3) is the Heston PDE, it also underpin the corresponding Hamilton-Jacobi-Bellman equation if I were to find an option price using the dynamic programming instead of the equivalent contingent claim analysis employed here. For the future reference, I'll use the so-called "Heston operator"

$$\mathcal{A} = (r - \delta)S_t \frac{\partial}{\partial S} + \frac{1}{2}v_t S_t^2 \frac{\partial^2}{\partial S^2} + (\kappa(\bar{v} - v_t) - \lambda v_t) \frac{\partial}{\partial v} + \frac{1}{2}\xi^2 v_t \frac{\partial^2}{\partial v^2} + \xi \rho v_t S_t \frac{\partial^2}{\partial S \partial v}$$

so that the PDE in the Equation (3): $(\frac{\partial}{\partial t} + \mathcal{A} - r)V = 0$. Mathematically, $\mathcal{A}$ is an elliptic (but degenerate) differential operator and the infinitesimal generator of the two-dimensional Itô process (S_t, v_t) under $\mathbb{Q}$. Intuitively, $\mathcal{A}$ defines the instantaneous drift of the option, such that $\mathbb{E}^{\mathbb{Q}}[dV_t] = (\frac{\partial}{\partial t} + \mathcal{A})V dt$. Risk-neutral pricing then imposes $(\frac{\partial}{\partial t} + \mathcal{A})V = rV$.

2 European Call Option Price

I first, with the help of Itô's Lemma, restate the Equation (3) in terms of $s = \ln S$ ⁵:

$$rV = \frac{\partial V}{\partial t} + \left(r - \delta - \frac{1}{2}v_t\right) \frac{\partial V}{\partial s} + \frac{1}{2}v_t \frac{\partial^2 V}{\partial s^2} + (\kappa(\bar{v} - v_t) - \lambda v_t) \frac{\partial V}{\partial v} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 V}{\partial v^2} + \xi \rho v_t \frac{\partial^2 V}{\partial s \partial v} \quad (4)$$

so that European call option becomes

$$\mathcal{C}(K) = e^{-r\tau} \mathbb{E}_t^{\mathbb{Q}}[(S_T - K)^+] \equiv e^{-\delta\tau} e^s P_1 - e^{-r\tau} K P_2 \quad (5)$$

where K is the strike price, $\tau = T - t$ is the time to maturity, P_1 and P_2 are exercise probabilities such that⁶

$$P_1 = \mathbb{Q}\{S_T > K\} \quad (6)$$

$$P_2 = \mathbb{Q}\{S_T > K\} \quad (7)$$

⁵To get here quick, apply Itô's Lemma to get to the SDE (19), then apply Feynman-Kac theorem described in the Section 5.2 to the SDE system with $\ell = (s_t - \ln K)^+$.

⁶This is Bakshi and Madan (2000) representation of call option, not exactly Heston (1993). However, the representation works for a wide range of models, so I show it here. For instance, those will be exactly $\mathcal{N}(d_1)$ and $\mathcal{N}(d_2)$ from the Black and Scholes (1973).

where $\mathbb{Q}' \sim \mathbb{Q}$ is the equivalent martingale measure that uses $S_t e^{\delta t}$ as numéraire. Specifically, the change in numéraire is $\frac{d\mathbb{Q}'}{d\mathbb{Q}}|_t = \frac{S_T e^{\delta T} / S_t e^{\delta t}}{e^{rT} / e^{rt}}$. Intuitively,

$$\mathbb{E}^{\mathbb{Q}}[S_T \mathbb{I}_{\{S_T > K\}}] = S_t e^{(r-\delta)\tau} \mathbb{Q}'\{S_T > K\}$$

There are two ways to derive P_1 and P_2 now. Heston (1993) offers the first way, which takes a bit of time:

1. Substitute Equation (5) into (4) – we proved the PDE must satisfy the derivative
2. Realise that equation (5) satisfies (4) with any $S_t \geq 0$, $K \geq 0$, $r \geq 0$ ⁷
3. Substitute $K = 0$, $S_t = 1$, and $\delta = 0$ into the result of substituting (5) into (4) to get the PDE for P_1 ⁸

$$\frac{\partial P_1}{\partial t} + \left(r - \delta + \frac{1}{2}v_t\right) \frac{\partial P_1}{\partial s} + \frac{1}{2}v_t \frac{\partial^2 P_1}{\partial s^2} + (\rho\xi v_t + \kappa(\bar{v} - v_t) - \lambda v_t) \frac{\partial P_1}{\partial v} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 P_1}{\partial v^2} + \xi\rho v_t \frac{\partial^2 P_1}{\partial s \partial v} = 0 \quad (8)$$

4. Substitute $K = 1$, $S_t = 0$, and $r = 1$ into the result of substituting (5) into (4) to get the PDE for $-P_2$ ⁹, and thus for P_2

$$\frac{\partial P_2}{\partial t} + \left(r - \delta - \frac{1}{2}v_t\right) \frac{\partial P_2}{\partial s} + \frac{1}{2}v_t \frac{\partial^2 P_2}{\partial s^2} + (\kappa(\bar{v} - v_t) - \lambda v_t) \frac{\partial P_2}{\partial v} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 P_2}{\partial v^2} + \xi\rho v_t \frac{\partial^2 P_2}{\partial s \partial v} = 0 \quad (9)$$

which yield the familiar Heston (1993) form:

$$\frac{\partial P_j}{\partial t} + (r - \delta - u_j v_t) \frac{\partial P_j}{\partial s} + \frac{1}{2}v_t \frac{\partial^2 P_j}{\partial s^2} + (a - b_j v_t) \frac{\partial P_j}{\partial v} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 P_j}{\partial v^2} + \xi\rho v_t \frac{\partial^2 P_j}{\partial s \partial v} = 0 \quad (10)$$

with $u_1 = \frac{1}{2}$ and $u_2 = -\frac{1}{2}$, $a = \kappa\bar{v}$, $b_1 = \kappa + \lambda - \rho\xi$ and $b_2 = \kappa + \lambda$.

Alternatively, I simply recognise that P_1 and P_2 are the exercise probabilities and thus must satisfy their respective PDEs – and apply Feynman-Kac theorem. Since $P_2 = \mathbb{E}^{\mathbb{Q}}[\mathbb{I}_{\{S_T > K\}}]$ ¹⁰, P_2 must satisfy the PDE (18) with $r(x, t) = 0$ (intuitively, pure probability is not discounted, it must be bounded between 0 and 1 at any point in time) and $f(x, t) = 0$ – this is the Equation . As for the P_1 , I go in the exact same manner. However, from Bakshi and Madan (2000) representation in the Equations (6), $P_1 = \mathbb{E}^{\mathbb{Q}'}[\mathbb{I}_{\{S_T > K\}}]$. I, therefore, first obtain the system of SDEs (21) and (22) under $\mathbb{Q}'$. Then P_1 must satisfy the PDE (18) with $r(x, t) = 0$ (again, pure probability is not discounted) and $f(x, t) = 0$. That is how the result in the Equation (8).

⁷Heston's argument.

⁸Values are such that to cancel $e^{-\delta\tau}e^s$.

⁹Values are such that to cancel $-e^{-r\tau}K$.

¹⁰i.e. $\ell = \mathbb{I}_{\{S_T > K\}}$

3 Characteristic Function

Clearly, the distribution of s_T does not admit an analytic form of the probability density function, characteristic function, $\varphi_j(\phi, s_t, v_t)$, is readily available. In fact, the characteristic functions $\varphi_1(\phi, s_t, v_t)$ and $\varphi_2(\phi, s_t, v_t)$ will follow the respective PDEs (8) and (2). What a lovely Feynman-Kac theorem! Given the definition of the characteristic function in the Equation (25), it must be true that characteristic functions would obey the respective PDEs under $\mathbb{Q}$ and $\mathbb{Q}'$. Heston (1993) conjectures

$$\varphi_j(\phi, s_t, v_t) = e^{C_j(\tau, \phi) + D_j(\tau, \phi)v_t + i\phi s_t}$$

substituting it to the PDE (10) yields

$$\begin{aligned} - \left(\frac{\partial C_j(\tau, \phi)}{\partial \tau} + v_t \frac{\partial D_j(\tau, \phi)}{\partial \tau} \right) + \rho \xi v_t i \phi D_j(\tau, \phi) - \frac{1}{2} v_t \phi^2 + \frac{1}{2} v_t \xi^2 D_j^2(\tau, \phi) \\ + (r - \delta + u_j v_t) i \phi + (a - b_j v_t) D_j(\tau, \phi) = 0 \end{aligned}$$

where $\tau = T - t$ ¹¹. I rearrange to have

$$\begin{aligned} v_t \left(- \frac{\partial D_j(\tau, \phi)}{\partial \tau} + \rho \xi i \phi D_j(\tau, \phi) - \frac{1}{2} \phi^2 + \frac{1}{2} \xi^2 D_j^2(\tau, \phi) + u_j i \phi - b_j D_j(\tau, \phi) \right) \\ - \frac{\partial C_j(\tau, \phi)}{\partial \tau} + (r - \delta) i \phi + a D_j(\tau, \phi) = 0 \end{aligned}$$

which Heston solves as a system of two differential equations:

$$\begin{cases} \frac{\partial D_j(\tau, \phi)}{\partial \tau} = \rho \xi i \phi D_j(\tau, \phi) - \frac{1}{2} \phi^2 + \frac{1}{2} \xi^2 D_j^2(\tau, \phi) + u_j i \phi - b_j D_j(\tau, \phi) \\ \frac{\partial C_j(\tau, \phi)}{\partial \tau} = (r - \delta) i \phi + a D_j(\tau, \phi) \end{cases} \quad (11)$$

The first equation in the system (11) is Riccati equation (27) with $p_j = u_j i \phi - \frac{1}{2} \phi^2$, $q_j = \rho \xi i \phi - b_j$, and $R = \frac{1}{2} \xi^2$. The section 8 offers a textbook solution to the general Riccati equation which I follow here for the Heston case. The corresponding second-order equation is, therefore,

$$\frac{\partial^2 w}{\partial \tau^2} - q_j \frac{\partial w}{\partial \tau} + p_j R w = 0$$

¹¹Because we describe the characteristic function for the s_T . note that $\frac{\partial f}{\partial t} = -\frac{\partial f}{\partial \tau}$.

with the roots to $x^2 - qx + pR = 0$ being

$$x_{1,2} = \frac{q_j \pm \sqrt{q_j^2 - 4p_jR}}{2} \equiv \frac{q_j \pm d_j}{2}$$

with $d_j = \sqrt{(b_j - \rho\xi i\phi)^2 - \xi^2(2u_j i\phi - \phi^2)}$. The known solution is

$$D_j(\tau, \phi) = -\frac{Mx_1e^{x_1t} + Nx_2e^{x_2t}}{Me^{x_1t} + Ne^{x_2t}} \frac{1}{R} \equiv -\frac{Lx_1e^{x_1t} + x_2e^{x_2t}}{Le^{x_1t} + e^{x_2t}} \frac{1}{R}$$

with $L = M/N$. To carry on I realise that at time T (i.e. at $\tau = 0$), $D_j(0, \phi) = 0$ ¹², meaning that $Lx_1 + x_2 = 0$.

Therefore, $L = x_1/x_2$. I further follow Heston (1993) notation¹³ to state

$$D_j(\tau, \phi) = \frac{b_j - \rho\xi i\phi - d_j}{\xi^2} \frac{1 - e^{-d_j\tau}}{1 - c_j e^{-d_j\tau}} \quad (12)$$

where $c_j = \frac{b_j - \rho\xi i\phi - d_j}{b_j - \rho\xi i\phi + d_j}$.

I find the solution to $C_j(\tau, \phi)$ by integrating the second equation in the system (11):

$$\begin{aligned} C_j(\tau, \phi) &= \int_0^\tau (r - \delta)i\phi du + av \frac{b_j - \rho\xi i\phi - d_j}{\xi^2} \int_0^\tau \frac{1 - e^{-d_j u}}{1 - c_j e^{-d_j u}} du \\ &= (r - \delta)i\phi\tau + a \frac{b_j - \rho\xi i\phi - d_j}{\xi^2} \int_0^\tau \frac{1 - e^{-d_j u}}{1 - c_j e^{-d_j u}} \frac{d_j e^{d_j u}}{d_j e^{d_j u}} du \\ &= (r - \delta)i\phi\tau + \frac{a}{d_j} \frac{b_j - \rho\xi i\phi - d_j}{\xi^2} \int_1^{e^{d_j\tau}} \frac{1 - e^{-d_j u}}{(1 - c_1 e^{-d_j u})e^{-d_j u}} d(e^{-d_j u}) \\ &= (r - \delta)i\phi\tau + \frac{a}{d_j} \frac{b_j - \rho\xi i\phi - d_j}{\xi^2} \int_1^{e^{d_j\tau}} \left[\frac{1}{e^{d_j u}} - \frac{1 - e^{-d_j u}}{1 - c_j e^{-d_j u}} \right] d(e^{-d_j u}) \\ &= (r - \delta)i\phi\tau + \frac{a}{\xi^2} \left((b_j - \rho\xi i\phi - d_j) - \ln \frac{1 - c_j e^{-d_j\tau}}{1 - c_j} \right) \end{aligned} \quad (13)$$

(e.g. long-run variance, mean reversion speed, volatility of volatility, and spot/variance correlation) to the market prices of a liquid set of vanilla options across strikes and maturities. Because the model offers a good forward-looking approximation the entire implied volatility surface dynamically, practitioners use it to price exotic options consistently with observed smiles while allowing for volatility randomness. It performs well for path-dependent products because it generates a realistic joint distribution of the underlying and variance over time. However, since the market is incomplete under stochastic volatility, perfect hedging is

¹²This is the terminal condition for the characteristic function, $\varphi_j(\phi, s_T, v_T) = \mathbb{E}_T[e^{i\phi s_T}] = e^{i\phi s_T}$. This means that $D_j(0, \phi) = C_j(0, \phi) = 0$.

¹³I use c_j as compared to Heston's g_j . The mapping is $c_j = 1/g_j$. It does not change anything, it is just more stable when implementing.

not possible: residual volatility risk remains even after continuous delta hedging, so hedging strategies often combine dynamic underlying hedges with static positions in vanillas to mitigate vega exposure.

4 Finite Difference Methods

Finite difference method is a numerical scheme to solve differential equations as, e.g., the Equation (3). The idea is to build a grid over the t , S , and v ¹⁴. Let me say we defined

$$S_i \in S_0, S_1, S_2, \dots, S_{N_S}$$

$$v_j \in v_0, v_1, v_2, \dots, v_{N_v}$$

$$t_n \in t_0, t_1, t_2, \dots, t_{N_t}$$

in whatever way we like¹⁵. $N_S + 1$, $N_v + 1$, and $N_t + 1$ are the number of the points in each grid with $S_0 = 0$, S_{N_S} being large (or rare) enough value at which options value does not change much or changes predictably (think of deep OTM and ITM options), $v_0 = 0$, v_{N_v} being large (or rare) enough to consider everything above irrelevant, $t_0 = 0$ (today), and t_{N_t} being maturity¹⁶.

4.1 Approximating the Derivatives

It is common to apply central differences to the first-order derivatives

$$\begin{aligned} \frac{\partial V(S_i, v_j, t_n)}{\partial S} &= \frac{V(S_{i+1}, v_j, t_n) - V(S_{i-1}, v_j, t_n)}{S_{i+1} - S_{i-1}} \\ \frac{\partial V(S_i, v_j, t_n)}{\partial v} &= \frac{V(S_i, v_{j+1}, t_n) - V(S_i, v_{j-1}, t_n)}{v_{j+1} - v_{j-1}} \end{aligned}$$

¹⁴The easiest way to go is to construct a uniform grid. This is okay, however, not always optimal – procedure can be rather computationally intensive. Moreover, with the uniform grid, the finer the grid the more accurate the result. Non-uniform are slightly more complicated but they can be lose in non-critical regions, and very fine in rather critical (e.g., near $S = K$, $v = 0$, $S = H$ where H is the barrier, $t = 0$ i.e. the day we value the option at, etc.).

¹⁵For vanilla options, see, e.g., In'T Hout and Foulon (2010) for the non-uniform grid that is naturally finer around strike and zero-variance.

¹⁶Generally, the bounds of the grid often correspond to boundary conditions of the PDE.

the second-order derivatives

$$\begin{aligned}\frac{\partial^2 V(S_i, v_j, t_n)}{\partial S^2} &= \frac{V(S_{i-1}, v_j, t_n)}{(S_i - S_{i-1})(S_{i+1} - S_{i-1})} - \frac{2V(S_i, v_j, t_n)}{(S_i - S_{i-1})(S_{i+1} - S_i)} + \frac{V(S_{i+1}, v_j, t_n)}{(S_{i+1} - S_i)(S_{i+1} - S_{i-1})} \\ \frac{\partial^2 V(S_i, v_j, t_n)}{\partial v^2} &= \frac{V(S_i, v_{j-1}, t_n)}{(v_j - v_{j-1})(v_{j+1} - v_{j-1})} - \frac{2V(S_i, v_j, t_n)}{(v_j - v_{j-1})(v_{j+1} - v_j)} + \frac{V(S_i, v_{j+1}, t_n)}{(v_{j+1} - v_j)(v_{j+1} - v_{j-1})}\end{aligned}$$

and, finally, mixed derivative¹⁷

$$\frac{\partial^2 V(S_i, v_j, t_n)}{\partial S \partial v} = \frac{V(S_{i+1}, v_{j+1}, t_n) - V(S_{i-1}, v_{j+1}, t_n) - V(S_{i+1}, v_{j-1}, t_n) + V(S_{i-1}, v_{j-1}, t_n)}{(S_{i+1} - S_{i-1})(v_{j+1} - v_{j-1})}$$

Importantly, that we must use the values of S_i and v_j that are only within the grid. Therefore, for the boundary values we can use either forward or backward differences (whatever relevant).

4.2 Boundary conditions

For the European call option, conditions are the following:

$$\begin{aligned}V(S_i, v_j, t_{N_t}) &= \max\{0, S_i - K\} \\ V(S_0, v_j, t_n) &= 0 \\ \frac{\partial V(S_i, v_0, t_n)}{\partial t} + (r - \delta)S_i \frac{\partial V(S_i, v_0, t_n)}{\partial S} + \kappa \bar{v} \frac{\partial V(S_i, v_0, t_n)}{\partial v} &= rV(S_i, v_0, t_n) \\ \frac{\partial V(S_{N_S}, v_j, t_n)}{\partial S} &= 1 \\ V(S_i, v_{N_v}, t_n) &= S_i\end{aligned}$$

4.3 Explicit Scheme

Explicit scheme is numerically simple, so I demonstrate it here:

$$V(S_i, v_j, t_{n-1}) = V(S_i, v_j, t_n) + dt(\mathcal{A}_{i,j} - r)V(S_i, v_j, t_n)$$

¹⁷See, e.g., Kluge (2002) for a more accurate expression. I here avoid it for the sake of space. Moreover, the expression perfectly works for uniform grids.

which is the special case of

$$\mathbf{V}(n-1) = (\mathbf{I} - \theta dt \mathbf{A})^{-1} (\mathbf{I} + (1 - \theta) dt \mathbf{L}) \mathbf{V}(n)$$

where $\mathbf{L}$ is the $(N_S + 1)(N_v + 1) \times (N_S + 1)(N_v + 1)$ square matrix based on $\mathcal{A} - r$. I won't go into details here, but I refer to Duffy (2006) for details¹⁸.

4.4 Practical Considerations: Log-Transform and Grid Construction

To improve numerical stability and ensure consistent discretisation, we perform a change of variables $s = \ln S$, transforming the PDE (??) into log-space. This eliminates multiplicative nonlinearity in S , making finite difference stencils more stable and uniform. The transformed Heston PDE becomes:

$$\frac{\partial V}{\partial t} = \frac{1}{2}v \frac{\partial^2 V}{\partial s^2} + \rho \xi v \frac{\partial^2 V}{\partial s \partial v} + \frac{1}{2}\xi^2 v \frac{\partial^2 V}{\partial v^2} + \left(r - \delta - \frac{1}{2}v\right) \frac{\partial V}{\partial s} + [\kappa(\theta - v) - \lambda v] \frac{\partial V}{\partial v} - rV$$

For the variance direction, a log-transform $y = \ln v$ is generally avoided due to the degeneracy of the CIR process near $v = 0$. Instead, a non-uniform grid is used in $v \in [0, v_{N_v}]$, dense near zero to resolve the singularity. Discretisation uses central differences in space and Crank–Nicolson or ADI time-stepping for second-order accuracy and unconditional stability.

4.5 Barrier Options and Path Dependence in Finite Differences

Finite difference methods can price barrier options by enforcing the barrier condition directly into the PDE domain. Consider a down-and-out European call with barrier H . In log-space, the barrier becomes $s_B = \ln H$. We define the spatial domain $s \in [s_B, s_{\max}]$, and enforce:

$$V(s_B, v, t) = 0, \quad (\text{absorbing barrier condition}). \quad (14)$$

For discretely monitored barriers, the barrier condition is imposed only at specified monitoring dates t_k , aligning the time grid with observation points. Grid refinement near the barrier enhances accuracy:

¹⁸See, however, how beautiful the expression encompasses various familiar schemes. For instance, $\theta = 0$ yield explicit scheme while $\theta = 1$ produces implicit scheme. Crank–Nicolson scheme is obtained with $\theta = \frac{1}{2}$.

- Use a non-uniform spatial grid with clustering near $s = s_B$.
- Time-stepping should preserve the barrier structure—early exercise or jumps are not allowed between monitoring steps.

Although barrier options are path-dependent, the finite difference method approximates their payoff by enforcing proper boundary behaviour. For continuous monitoring, this yields accurate results. However, for complex path-dependent structures (e.g., Asian, lookback), Monte Carlo methods remain more natural due to their trajectory-based modelling.

For an up-and-in European call option with barrier H , we define the log-barrier as $s_B = \ln H$. Since the payoff activates upon crossing the barrier, the domain must include s_B , and activation must be enforced during evolution.

For continuously monitored barriers, we apply the in-out parity:

$$V^{\text{up-in}}(s, v, t) = V^{\text{vanilla}}(s, v, t) - V^{\text{up-out}}(s, v, t), \quad (15)$$

where $V^{\text{up-out}}$ satisfies the absorbing boundary condition

$$V^{\text{up-out}}(s_B, v, t) = 0.$$

For discretely monitored barriers, we define monitoring times $\{t_k\}$ and update values at each t_k as

$$V(s, v, t_k^-) = \begin{cases} V(s, v, t_k^+), & \text{if } s < s_B \\ V^{\text{vanilla}}(s, v, t_k^+), & \text{if } s \geq s_B. \end{cases}$$

A non-uniform grid in s is recommended, clustering points near s_B , to accurately capture activation dynamics. Time steps should align with monitoring dates to ensure barrier detection.

5 Variance Risk Premium

In this section, I try a different method in deriving the PDE. I first change the probability measure from $\mathbb{P}$ to $\mathbb{Q}$, then I apply Feynman-Kac theorem. Perhaps, the approach, although less economically motivated rather than mathematically, may offer some intuition into how come we make a substitution with (2).

5.1 Change in Probability Measure (from $\mathbb{P}$ to $\mathbb{Q}$)

Let me first restate the SDEs:

$$\begin{aligned} dS_t &= \alpha S_t dt + \sqrt{v_t} S_t \sqrt{1 - \rho^2} dW_t^{\iota, \mathbb{P}} + \sqrt{v_t} S_t \rho dW_t^{v, \mathbb{P}} \\ dv_t &= \kappa(\bar{v} - v_t) dt + \xi \sqrt{v_t} dW_t^{v, \mathbb{P}} \end{aligned}$$

where $dW_t^{\iota, \mathbb{P}}$ is the independent Brownian motion so that $dW_t^{S, \mathbb{P}} = \sqrt{1 - \rho^2} dW_t^{\iota, \mathbb{P}} + \rho dW_t^{v, \mathbb{P}}$.

Under the measure $\mathbb{Q}$, asset's expected return equals the risk-free rate of return (minus the cashflows asset generates). However, since market is not complete (variance is not traded) we manually specify the variance risk premium. Heston (1993) assumes it to be linear in variance, so with Girsanov's theorem:

$$\begin{aligned} dW_t^{v, \mathbb{Q}} &= dW_t^{v, \mathbb{P}} + \frac{\lambda v_t}{\xi \sqrt{v_t}} dt \\ dW_t^{\iota, \mathbb{Q}} &= dW_t^{\iota, \mathbb{P}} + \frac{\mu - r - \frac{\rho}{\xi} \lambda v_t}{\sqrt{v_t} \sqrt{1 - \rho^2}} dt \end{aligned}$$

with $\mu = \alpha + \delta$, so that under $\mathbb{Q}$

$$\begin{aligned} dS_t &= (r - \delta) S_t dt + \sqrt{v_t} S_t \sqrt{1 - \rho^2} dW_t^{\iota, \mathbb{Q}} + \sqrt{v_t} S_t \rho dW_t^{v, \mathbb{Q}} \\ dv_t &= (\kappa(\bar{v} - v_t) - \lambda v_t) dt + \xi \sqrt{v_t} dW_t^{v, \mathbb{Q}} \end{aligned}$$

or, in a more familiar Heston-form,

$$dS_t = (r - \delta) S_t dt + \sqrt{v_t} S_t dW_t^{S, \mathbb{Q}} \tag{16}$$

$$dv_t = (\kappa(\bar{v} - v_t) - \lambda v_t) dt + \xi \sqrt{v_t} dW_t^{v, \mathbb{Q}} \tag{17}$$

Importantly, since the market is not complete, measure $\mathbb{Q}$ is not unique – and subject to λ estimation¹⁹

5.2 Feynman-Kac Theorem

Heston (1993) suggests Equating (2) with the PDE of V , to get to the Equation (3). Quite a leap! However, there is intuition. While $(r - \delta)$ is the drift of the S_t under measure $\mathbb{Q}$, $\kappa(\bar{v} - v_t) - \lambda(S, v, t)$ represents the

¹⁹This is the instance of the Second Fundamental Theorem of Asset Pricing. In fact, we do not have to specify the variance risk premium as λv_t , in case it makes sense – see, e.g., Bollerslev et al. (2011).

“drift” of the variance under the measure $\mathbb{Q}$ ²⁰. In fact, applying Feynman-Kac theorem to the SDEs (16) and (17) yields exactly (3). In its essence, the Feynman-Kac theorem states that if we have the PDE of the form

$$\frac{\partial V}{\partial t} + \sum_{i=1}^N \mu_i(x, t) \frac{\partial V}{\partial x_i} + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \sigma_{i,j}(x, t) \frac{\partial^2 V}{\partial x_i \partial x_j} - r(x, t)V = h(x, t) \quad (18)$$

with the terminal condition $V(x, T) = \ell(x)$, and the process $X_t \in \mathbb{R}^N$ satisfies

$$dX_t = \mu(X_t, t)dt + \Sigma(X_t, t)dW_t^{\mathbb{Q}}$$

then the solution is given by

$$V(x, t) = \mathbb{E}^{\mathbb{Q}} \left[\int_t^T e^{-\int_t^T r(X_\tau, \tau) d\tau} f(X_u, u) du + e^{-\int_t^T r(X_\tau, \tau) d\tau} \ell(X_T) \right]$$

so that in case of Heston (1993) model $X_t = \begin{bmatrix} S_t \\ v_t \end{bmatrix}$, $\mu_t = \begin{bmatrix} (r - q)S_t \\ \kappa((\bar{v} - v_t) - \lambda v_t) \end{bmatrix}$, and $\Sigma_t = \sqrt{v_t} \begin{bmatrix} \rho S_t & \sqrt{1 - \rho^2} S_t \\ \xi & 0 \end{bmatrix}$,

assuming $dW_t^{\mathbb{Q}} = \begin{bmatrix} dW_{1,t}^{\mathbb{Q}} \\ dW_{2,t}^{\mathbb{Q}} \end{bmatrix}$. Also, $r(x, t) = r$, $h(x, t) = 0$, and $\ell(x) = (S_T - K)^+$.

Feynman-Kac theorem alone is not enough to price options, Feynman-Kac theorem with the equivalent martingale measure $\mathbb{Q}$ is²¹. The First Fundamental Theorem of Asset Pricing states that existence of measure $\mathbb{Q}$ implies the PDE (3) does not admit arbitrage. Therefore, PDE is the option-pricing PDE. This is the modern way of deriving Heston (1993) PDE.

6 Change in Measure (from $\mathbb{Q}$ to $\mathbb{Q}'$)

Let me first take the system of SDEs in the Equation (16) and (17) and apply Itô's Lemma to $\ln S_t = s_t$:

$$ds_t = \left(r - \delta - \frac{1}{2} v_t \right) dt + \sqrt{v_t} dW_t^{S, \mathbb{Q}} \quad (19)$$

$$dv_t = (\kappa(\bar{v} - v_t) - \lambda v_t) dt + \xi \sqrt{v_t} dW_t^{v, \mathbb{Q}} \quad (20)$$

²⁰Clearly, change measure is effectively changing the drift.

²¹Note that Feynman-Kac theorem holds for any probability measure, we require risk-neutral measure for our results to make economic sense.

Next, I apply the change in numéraire so that to express the processes under the measure $\mathbb{Q}' \sim \mathbb{Q}$, the equivalent martingale measure that uses $S_t e^{\delta t}$ as numéraire. The change in numéraire is

$$\frac{d\mathbb{Q}'}{d\mathbb{Q}}|_t = \frac{S_T e^{\delta T} / S_t e^{\delta t}}{e^{rT} / e^{rt}}$$

so that the new Brownian motions are²²

$$\begin{aligned} dW_t^{S, \mathbb{Q}'} &= dW_t^{S, \mathbb{Q}} - \sqrt{v_t} dt \\ dW_t^{v, \mathbb{Q}} &= dW_t^{v, \mathbb{Q}} - \rho \sqrt{v_t} dt \end{aligned}$$

and I substitute them in the Equations (19) and (20) to get

$$ds_t = \left(r - \delta + \frac{1}{2} v_t \right) dt + \sqrt{v_t} dW_t^{S, \mathbb{Q}'} \quad (21)$$

$$dv_t = (\rho \xi v_t + \kappa(\bar{v} - v_t) - \lambda v_t) dt + \xi \sqrt{v_t} dW_t^{v, \mathbb{Q}'} \quad (22)$$

7 Semi-Analytic Probability Expression

Gil-Pelaez (1951) derives the expression for the probability in terms of the characteristic function:

$$\mathbb{P}\{s_T > k\} = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iuk} \varphi(u)}{iu} du \right] \quad (23)$$

which is obtained as the Equation (26):

$$\mathbb{P}\{s_T > k\} = 1 - F_s(k) \equiv \int_k^\infty f(x) dx = \frac{1}{2\pi} \int_k^\infty \left(\int_{-\infty}^\infty e^{-iux} \varphi(u) du \right) dx \equiv \frac{1}{2\pi} \int_{-\infty}^\infty \varphi(u) \left(\int_k^\infty e^{-iux} dx \right) du$$

I refer reader to a very elegant single-page solution resulting in (23) in Gil-Pelaez (1951). Intuitively, however, it is just as if we have the pdf and cdf.

²²Girsanov's theorem tells us effectively how to adjust the drift: $dW_t^{\mathbb{Q}} + \vartheta_t dt$. Usually, we have a market price of risk as ϑ_t . Here, however, we have already adjusted from market price of risk when switched from $\mathbb{P}$ to $\mathbb{Q}$. Note that $\frac{d\mathbb{Q}'}{d\mathbb{Q}}|_t \propto \frac{S_T}{S_t} \equiv e^{\ln S_T - \ln S_t}$ (I ignore deterministic part here). Therefore, the change of measure is about $\ln S_T - \ln S_t$. Therefore,

$$\frac{d\mathbb{Q}'}{d\mathbb{Q}}|_t \propto e^{\int_t^T (r - \delta - \frac{1}{2} v_u) du + \int_t^T \sqrt{v_u} dW_u^{S, \mathbb{Q}}}$$

so that to have s_t following martingale under $\mathbb{Q}'$, I need to subtract $\sqrt{v_t} dt$ (the coefficient next to the $dW_u^{S, \mathbb{Q}}$).

Then, since $\mathbb{E}^{\mathbb{Q}}[dW_t^{S, \mathbb{Q}} dW_t^{v, \mathbb{Q}}] = \rho dt$, I shift the drift of v_t by $-\rho \sqrt{v_t} dt$.

7.1 A Note on the Fourier Transform

In finance we define the Fourier Transform of a function f to be

$$\tilde{f}(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx \quad (24)$$

since it is convenient when $f(x)$ is the pdf of the random variable X – in this case the Fourier transform is the characteristic function for X ²³:

$$\varphi(u) = \mathbb{E}[e^{iux}] \equiv \tilde{f}(u) \quad (25)$$

The pdf is recovered via the inverse Fourier transform:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} \varphi(u) du \quad (26)$$

8 A Note on Riccati Equation

The Riccati equation for some function $f(t)$ with coefficients $p(t)$, $q(t)$, and $r(t)$ is defined as

$$\frac{\partial f(t)}{\partial t} = p(t) + q(t)f(t) + R(t)f^2(t) \quad (27)$$

the textbook solution for which is through the second order differential equation for $\omega(t)$:

$$\frac{\partial^2 \omega(t)}{\partial t^2} - \left(\frac{1}{p(t)} \frac{\partial p(t)}{\partial t} + q(t) \right) \frac{\partial \omega(t)}{\partial t} + p(t)R(t)\omega(t) = 0$$

yielding the solution

$$f(t) = -\frac{1}{\omega(t)} \frac{\partial \omega(t)}{\partial t} \frac{1}{R(t)}$$

so that the problem reduces to solving for $\omega(t)$ by solving $x^2 + bx + c = 0$ ²⁴ which is a middle school problem on solving second-order polynomial:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4c}}{2}$$

²³If a random variable admits a probability density function, then the characteristic function is the Fourier transform (with sign reversal) of the probability density function. Thus it provides an alternative route to analytical results compared with working directly with probability density functions or cumulative distribution functions – very convenient when the pdf and/or cdf is not available, for instance, as in the case of S_T in the Heston (1993).

²⁴Let $a = 1$, $b = \frac{1}{p(t)} \frac{\partial p(t)}{\partial t} + q(t)$, and $c = p(t)R(t)$.

and substituting it in a known-form solution

$$\omega(t) = Me^{x_1 t} + Ne^{x_2 t}$$

with M and N being constants. Finally,

$$f(t) = -\frac{Mx_1e^{x_1 t} + Nx_2e^{x_2 t}}{Me^{x_1 t} + Ne^{x_2 t}} \frac{1}{R(t)}$$